enough to encompass today's observable universe looks very nearly flat. This is a feature of geometry we are all well aware of: The surface of a basketball is obviously curved, but it took both time and thinkers with chutzpah before everyone was convinced that the earth's surface was also curved. The reason is that, all else being equal, the larger something is, the more gradually it curves and the flatter a patch of a given size on its surface appears. If you draped the state of Nebraska over a sphere just a few hundred miles in diameter, as in Figure 10.4a, it would look curved, but on the earth's surface, as just about all Nebraskans concur, it looks flat. If you laid Nebraska out on a sphere a billion times larger than earth, it would look flatter still. In inflationary cosmology, space was stretched by such a colossal factor that the observable universe, the part we can see, is but a small patch in a gigantic cosmos. And so, like Nebraska laid out on a giant sphere as in Figure 10.4d, even if the entire universe were curved, the *observable* universe would be very nearly flat.[18]

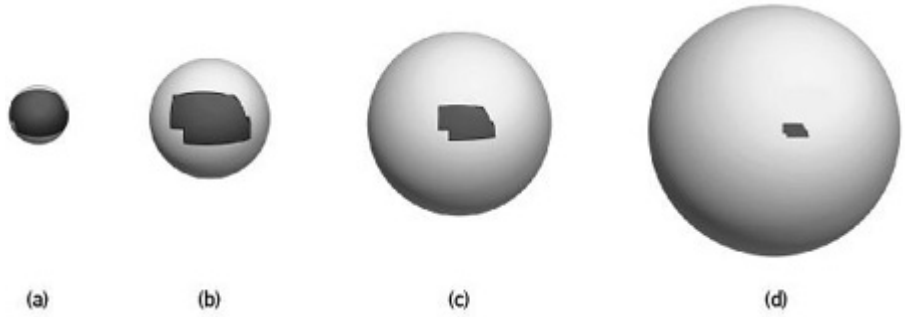


Figure 10.4 A shape of fixed size, such as that of the state of Nebraska, appears flatter and flatter when laid out on larger and larger spheres. In this analogy, the sphere represents the entire universe, while Nebraska represents the *observable* universe—the part within our cosmic horizon.

It's as if there are powerful, oppositely oriented magnets embedded in the mountain climber's boots and the thin ledge she is crossing. Even if her step is aimed somewhat off the mark, the strong attraction between the magnets ensures that her foot lands squarely on the ledge. Similarly, even if the early universe deviated a fair bit from the critical matter/energy density and hence was far from flat, the inflationary expansion ensured that the part of space we have access to was *driven* toward a flat shape and that the matter/energy density we have access to was *driven* to the critical value.